

Partial Differential Equation:

A partial differential equation (PDE) is an equation involving one or more partial derivatives of an unknown function of several variables. If $z = f(x, y)$ then

$$\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0 \text{ is a PDE.}$$

The order of a PDE is the order of the highest derivative that appears in the equation.

$$\text{Example: } \frac{\partial z}{\partial x} + \frac{\partial^2 z}{\partial y^2} = 0 \quad [\text{order 2}]$$

The degree of such equation is the greatest exponent of the highest order.

$$\text{Example: } \frac{\partial z}{\partial x} + \left(\frac{\partial^2 z}{\partial y^2} \right)^2 = 0 \quad [\text{degree 2}]$$

The partial differential equation

$$F(x, y, z, t; u, u_x, u_y, u_z, u_t, u_{xx}, u_{yy}, \dots) = 0$$

is said to be linear if the function F is algebraically linear in each variables $u, u_x, u_y, \dots$, and if the coefficients of u and its derivatives are functions only of independent variables.

An equation that is not linear is said to be nonlinear; a nonlinear equation is quasilinear if it is linear in the highest derivative.

Examples: ① $\frac{\partial z}{\partial x} + \frac{\partial^2 z}{\partial y^2} + z = 0$ (linear)

② $\left(\frac{\partial z}{\partial x}\right)^2 + \frac{\partial^2 z}{\partial y^2} = 0$ (Non-linear)

Notation:

When we consider the case of two independent variables we usually assume to be x and y and assume z to be the dependent variables. It is customary to represent $\frac{\partial z}{\partial x}$ by p and $\frac{\partial z}{\partial y}$ by q . Also we write

$$r = \frac{\partial^2 z}{\partial x^2}, \quad s = \frac{\partial^2 z}{\partial x \partial y}, \quad t = \frac{\partial^2 z}{\partial y^2}.$$

In case there are n independent variables, we take them to be $x_1, x_2, \dots, x_n$ and z is regarded as the dependent variables. In this case, we use the following notations:

$$p_1 = \frac{\partial z}{\partial x_1}, \quad p_2 = \frac{\partial z}{\partial x_2}, \quad \dots \quad p_n = \frac{\partial z}{\partial x_n}.$$

Sometimes the partial differentiations are also denoted by making use of suffixes. Thus we write

$$u_x = \frac{\partial u}{\partial x}, \quad u_y = \frac{\partial u}{\partial y}, \quad u_{xx} = \frac{\partial^2 u}{\partial x^2}, \quad u_{xy} = \frac{\partial^2 u}{\partial x \partial y} \text{ and}$$

so on.

Origin of Partial Differential Equations:

We shall now examine the interesting question of how partial differential equations arise. We show that such equations can be formed by the elimination of arbitrary constants or arbitrary functions.

The partial differential equations may be derived in two ways:

- I. By the elimination of arbitrary constants from a given relation between the variables.
- II. By the function elimination of arbitrary functions of the variables.

Elimination of arbitrary constants:

Let the relation connecting x, y and z be

$$F(x, y, z, a, b) = 0 \quad \text{--- (1)}$$

in which a and b are arbitrary constants. Let z be regarded as function of two independent variables

x and y . Differentiating ① with respect to x and y partially in turn, we get

$$\left. \begin{aligned} \frac{\partial F}{\partial x} + p \frac{\partial F}{\partial y} &= 0 \\ \text{and } \frac{\partial F}{\partial y} + q \frac{\partial F}{\partial z} &= 0 \end{aligned} \right\} \text{--- ②}$$

Eliminating two constants a and b from three equations of ① and ②, we shall obtain an equation of the form

$$f(x, y, z, a, b) = 0 \text{ --- ③}$$

which is the partial differential equation of first order.

Note:

Here we have two independent variables and two arbitrary constants, and the result of the elimination of the constants is a partial differential equation of the first order.

If however, the number of constants be greater than the number of independent variables, then the result of the elimination of the constants is usually a partial differential equation of order higher than the first.

Problem 1: Eliminate the arbitrary constants a and b from

$$z = ax^2 + by^2 + ab.$$

Solution: Given,

$$z = ax^2 + by^2 + ab \quad \text{--- ①}$$

Differentiating ① with respect to x and y we have

$$\frac{\partial z}{\partial x} = 2ax \Rightarrow p = 2ax$$

$$\text{and } \frac{\partial z}{\partial y} = 2by \Rightarrow q = 2by$$

Eliminate a and b we get

$$z = \left(\frac{p}{2x}\right)x^2 + \left(\frac{q}{2y}\right)y^2 + \frac{p}{2x} \cdot \frac{q}{2y}$$

$$\Rightarrow pq + 2px^2y + 2qxy^2 = 4xyz$$

which is the required partial differential equation of order one.

Problem 2: Eliminate the arbitrary constants a, b, c from

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

Solution: Here we have three arbitrary constants and two independent variables; hence the result will be a partial differential equation of order higher than the first.

Given that,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad \text{--- (1)}$$

Differentiating partially with respect to x and y , we have

$$\frac{2x}{a^2} + \frac{2z}{c^2} \cdot \frac{\partial z}{\partial x} = 0$$

$$\Rightarrow \frac{2x}{a^2} + \frac{2z}{c^2} p = 0 \quad \text{--- (2)}$$

$$\text{And } \frac{2y}{b^2} + \frac{2z}{c^2} \cdot \frac{\partial z}{\partial y} = 0$$

$$\Rightarrow \frac{2y}{b^2} + \frac{2z}{c^2} q = 0 \quad \text{--- (3)}$$

We can differentiate the equation (2) with respect to x and or the equation (3) with respect to y , to get another relation necessary for the elimination of the three constants.

Let us first differentiate (2) with respect to x , then

$$\frac{2}{a^2} + \frac{2}{c^2} \left[\frac{\partial z}{\partial x} \cdot p + z \cdot \frac{\partial^2 z}{\partial x^2} \right] = 0$$

$$\Rightarrow \frac{1}{a^2} + \frac{1}{c^2} [p^2 + zr] = 0$$

$$\Rightarrow p^2 + zr = -\frac{c^2}{a^2}$$

$$\Rightarrow p^2 + zr = \frac{z}{x} p \quad [\text{from (2)}]$$

$$\Rightarrow xp^2 + xzr - zp = 0$$

which is the required partial differential equation of order two.

Similarly, differentiating (3) with respect to y , and eliminating the constants, we get

$$yq^2 + yzt - zq = 0$$

a partial differential equation of second order.

Problem 3: Eliminate a, b from the eqⁿ: $z = ax + by + ab$

Solution:

Given $z = ax + by + ab$ — (1)

• Differentiating (1) partially w.r. to x and y , we get

$$\frac{\partial z}{\partial x} = a \Rightarrow p = a$$

$$\text{and } \frac{\partial z}{\partial y} = b \Rightarrow q = b$$

Putting the value of a & b in (1), we get

$$z = px + qy + pq$$

which is the required PDE of order one.

Problem 4:

Eliminate a and b from

$$z = ax + a^2y^2 + by$$

Solution:

Given $z = ax + a^2y^2 + by$ — (1)

Differentiating (1) partially w.r. to x and y , we get

$$\frac{\partial z}{\partial x} = a \Rightarrow p = a \quad \text{--- (2)}$$

$$\text{and } \frac{\partial z}{\partial y} = 2a^2y + b$$

$$\Rightarrow q = 2p^2y + b \quad ; [\text{by (1)}]$$

$$\Rightarrow b = q - 2p^2y$$

Putting the values of a and b in (1), we get

$$Z = px + p^2 y^2 + (a - 2p^2 y)y$$

$$= px + p^2 y^2 + ay - 2p^2 y^2$$

$$Z = px - p^2 y^2 + ay$$

which is the required PDE of order one.

Problem 5:

Eliminate a, b from

$$Z = (x^2 + a)(y^2 + b)$$

Solution:

$$\text{Given } Z = (x^2 + a)(y^2 + b) \quad \text{--- (1)}$$

Differentiating (1) partially w.r. to x and y , we get

$$\frac{\partial Z}{\partial x} = 2x(y^2 + b)$$

$$\Rightarrow p = 2xy^2 + 2xb$$

$$\Rightarrow \frac{p}{2x} = y^2 + b$$

$$\therefore b = \frac{p}{2x} - y^2$$

$$\text{and } \frac{\partial Z}{\partial y} = (x^2 + a)2y$$

$$\Rightarrow a = (x^2 + a)2y$$

$$\Rightarrow \frac{a}{2y} = x^2 + a$$

$$\therefore a = \frac{a}{2y} - x^2$$

Putting the values of a and b in (1) we get

$$Z = (x^2 + \frac{a}{2y} - x^2)(y^2 + \frac{p}{2x} - y^2) = \frac{a}{2y} \cdot \frac{p}{2x}$$

$\therefore p a = 4xyZ$ which is the required PDE.

Problem 6:

Eliminate a and b from the following equation of the form

$$(x-a)^2 + (y-b)^2 + z^2 = c^2$$

Solution:

Given the equation

$$(x-a)^2 + (y-b)^2 + z^2 = c^2 \quad \text{--- (1)}$$

Differentiating (1) w.r. to x and y , we have,

$$2(x-a) + 2z \frac{\partial z}{\partial x} = 0$$

$$\Rightarrow x-a + z p = 0$$

$$\therefore a = x + z p$$

$$\text{and } 2(y-b) + 2z \frac{\partial z}{\partial y} = 0$$

$$\Rightarrow y-b + z q = 0$$

$$\therefore b = y + z q$$

Putting the value of a and b in (1), we have

$$(x-x-zp)^2 + (y-y-zq)^2 + z^2 = c^2$$

$$\Rightarrow z^2 p^2 + z^2 q^2 + z^2 = c^2$$

$$\Rightarrow z^2 (p^2 + q^2 + 1) = c^2$$

which is the required PDE.

Problem 7:

Eliminate a, b, c from

$$z = ax + by + cxy \quad \text{and form the PDE}$$

Solution:

Given $z = ax + by + cxy$ — (1)

∴ Differentiating (1) w.r. to x and y , we get

$$\frac{\partial z}{\partial x} = a + cy$$

$$\Rightarrow cy + a - \frac{\partial z}{\partial x} = 0 \quad \text{--- (2)}$$

$$\frac{\partial z}{\partial y} = b + cx$$

$$\Rightarrow cx + b - \frac{\partial z}{\partial y} = 0 \quad \text{--- (3)}$$

Now differentiating (2) w.r. to x and (3) w.r. to y , we get

$$\frac{\partial^2 z}{\partial x^2} = 0 \quad \text{i.e. } r = 0$$

$$\text{and } \frac{\partial^2 z}{\partial y^2} = 0 \quad \text{i.e. } t = 0$$

Again differentiating (2) w.r. to y and (3) w.r. to x , we get

$$c - \frac{\partial^2 z}{\partial y \partial x} = 0$$

$$\Rightarrow c = \frac{\partial^2 z}{\partial y \partial x}$$

$$\therefore c = r$$

∴ From (2) we get

$$a = \frac{\partial z}{\partial x} - cy = p - ry$$

and from (3)

$$b = \frac{\partial z}{\partial y} - cx = q - rx$$

Putting the value of a, b, c in (1), we get

$$z = (p - ry)x + (av - rx)y + rxy$$

$$\therefore z = px + ay + rxy$$

which is the required PDE.

class
Problem 8:

Eliminate a and b from the following

$$z = (x+a)(y+b)$$

Solution:

$$\text{Given } z = (x+a)(y+b) \quad \text{--- (1)}$$

Differentiating (1) partially w.r. to x and y , we get

$$\frac{\partial z}{\partial x} = y+b$$

$$\Rightarrow p = y+b$$

$$\therefore b = p - y$$

$$\frac{\partial z}{\partial y} = x+a$$

$$\Rightarrow v = x+a$$

$$\therefore a = v - x$$

Putting the values of a and b in (1), we get

$$z = (x + v - x)(y + p - y)$$

$$\therefore z = pv$$

which is the required PDE.